



Vu1nLab

A Gamified Cybersecurity
Learning Platform

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Research Question

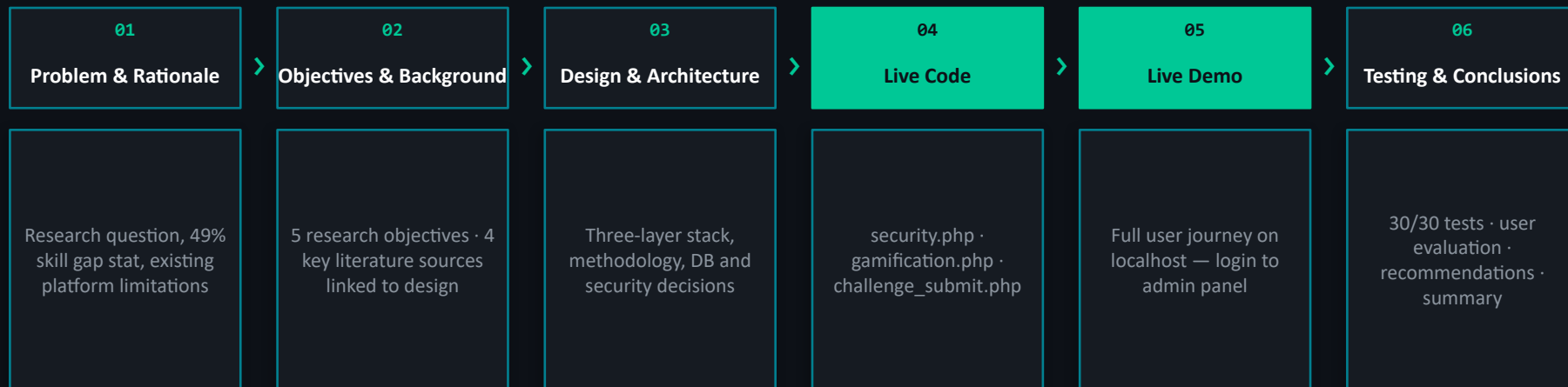
To what extent can a browser-based gamified learning platform improve accessibility, engagement and practical understanding of cybersecurity concepts for beginner learners?

Module: CSY4022 · Supervisor: Dr Mandy Morell

Word count: 14,892 · April 2026

Introduction — Viva Structure

This viva presents VulnLab — a browser-based gamified cybersecurity learning platform built to address a demonstrable skills gap in the UK workforce. The viva is structured in the following sections:



Sections 04 & 05 are live — slides minimised during those sections.

Problem & Rationale

49%

UK businesses report a
technical cybersecurity skill gap

DSIT, 2025

The Gap

- Learners understand theory but can't apply it in practice
- TryHackMe / HackTheBox assume prior knowledge and need VPN + VMs
- Technical setup overhead blocks beginners before they start
- No structured, browser-based, beginner-safe alternative exists

VulnLab Solution

Browser-based · No setup required · Locally simulated · Structured progression · Gamified · Ethically safe

5 Critical Success Factors defined from the literature:

Accessibility

Structured Progression

Gamified Engagement

Practical Interaction

Ethical Safety

Objectives & Background Research

Research Objectives

Obj 1	Identify limitations of existing platforms
Obj 2	Define Critical Success Factors
Obj 3	Design & develop VulnLab
Obj 4	Evaluate via testing + user feedback
Obj 5	Recommend best practices

Key Literature

Sweller (1988) — Cognitive Load Theory

Too much at once impairs learning. Drove skill-locking, hints, staged progression.

Ryan & Deci (2000) — Self-Determination Theory

Motivation from autonomy, competence, relatedness. Drove XP, unlocks, achievements.

Dicheva et al. (2015) — Gamification in Education

Rewards work when they reinforce learning. Informed how gamification was layered in.

Chothia & Novakovic (2015) — CTF-Style Challenges

Hands-on challenges help but overwhelm beginners. Drove hints and structure.

Design & Architecture

Three-Layer Architecture

Presentation Layer

HTML · CSS · JavaScript



Application Layer

PHP · security.php · layout.php · gamification.php



Data Layer

MySQL · 9 normalised tables · XAMPP

Modular codebase

Each concern lives in its own file — auth, dashboard, courses, challenges, gamification, tournaments, admin. A central layout.php and security.php apply consistently across every page via require_once.

Key Design Decisions

UNIQUE KEY on user_solves (user_id, challenge_id)

Prevents double XP at the database level — not just in application code.

Separate replay tables

user_course_restarts + user_replay_solves — replay without losing original XP.

XP never stored – always calculated

getUserScore() sums from user_solves live. Adjustment column handles admin rank changes.

Agile / iterative methodology

Build → test → refine per feature. Waterfall rejected — requirements evolved. (Sommerville, 2016)

Testing & Evaluation

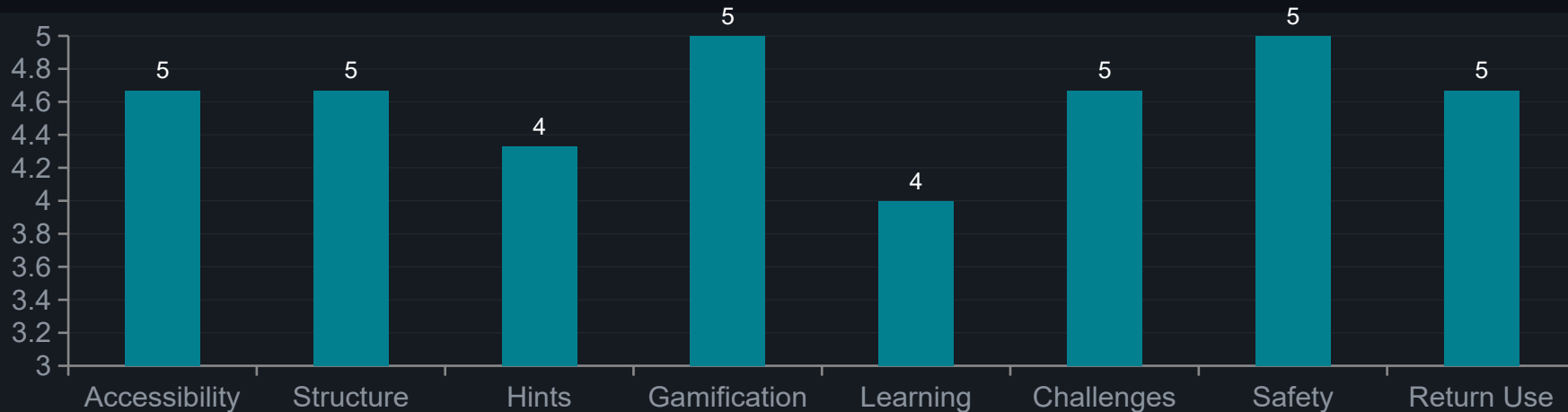
30 / 30

Tests Passed

Key Test Cases (Chapter 8 – full table of 30)

- | | |
|--------------------------|---|
| ✓ CSRF token validation | POST without token rejected — 'Invalid request' |
| ✓ Rate limiting | Account locked after 5th failed login |
| ✓ No double XP on replay | Replay mode prevented re-award |
| ✓ Skill-level locking | Beginner user denied /sqli direct URL |

User Evaluation (3 beginner participants • avg 4.63 / 5 • Chapter 8)



Conclusions & Recommendations

What Was Achieved

- All 5 objectives met
- 30 / 30 tests passed
- Average user score 4.63 / 5
- Full gamification — XP, gems, streaks, 23 achievements
- OWASP-aligned security implementation

5 Best-Practice Recommendations

- Accessibility first — remove the setup barrier
- Skill-lock content to manage cognitive load
- Gamification must reinforce, not distract
- Hints should guide, not reveal answers
- Simulated environments keep it ethically safe

Limitations & Future Work

- Only 3 participants — familiarity bias
- JS simulations ≠ real tools (Burp, Nmap)
- Local only — no long-term usage data
- Future: cloud deploy + larger user study
- Future: more network challenges

Summary

Research Question	To what extent can a browser-based gamified platform improve accessibility, engagement and cybersecurity learning for beginners?
Approach	Full-stack PHP/MySQL platform built iteratively using OWASP security standards, Cognitive Load Theory, and Self-Determination Theory.
Key Outcomes	All 5 objectives met · 30/30 tests passed · Avg user score 4.63/5 · 5 best-practice recommendations defined.
Limitations	Small user sample (3 participants), local-only deployment, JS simulations not equivalent to real security tools.
Future Work	Cloud deployment, larger longitudinal study, additional network and forensics challenges, real tool integration.